



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic: ...Algebra Of Events.....

LECTURE -

EXPLANATION

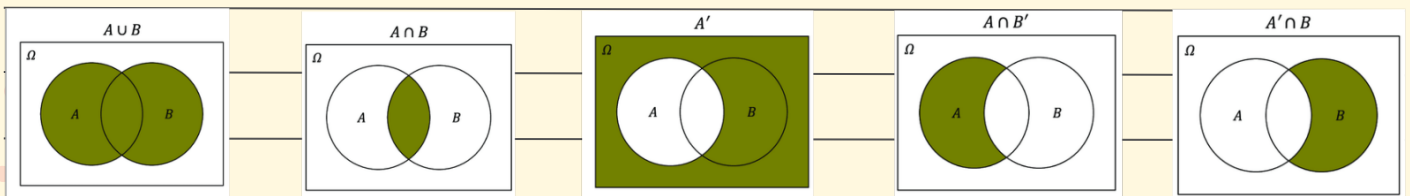
Event A or B : Union of event A and B. ( $A \cup B / A \vee B / A+B$ )

Event A and B : Intersection of event A and B. ( $A \cap B / A \wedge B / A \cap B$ )

Complementary Event : "not A" ( $A' / \bar{A} / A^c$ )

Event A but not B : Output of event A but not B ( $A - B / A \cap B'$ )

Event B but not A : Output of event B but not A ( $B - A / A' \cap B$ )



Random Drawn (Two or more) → Use combination

Example - A bag contain 3 Red & 4 black balls. Two balls are drawn at the random. Find the probability of getting  
i) Both red balls (ii) Both black balls (iii) One red and one black ball.

Addition Law -

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

In case of mutually exclusive events A, B & C

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

NOTES PREPARED BY: ANKIT ARORA